

CS 4134 – Quantum Computation and Information Processing

Exam 5

Virginia Tech Department of Computer Science, Spring 2026
Instructor: Sumeet Khatri

Name:

Date: Tuesday, May 5, 2026

Number of problems: 6

Time allowed: 75 minutes

Instructions: Please show all of your work. Your grade will be based not only on writing down the correct answer, but having a complete solution.

Grading: You will start from a grade of 100%. For every missed step, unclear or unexplained step, and mistake, you will lose two points, with a maximum of 10 points of deduction per problem. Unattempted problems result in a 10-point deduction.

Potentially helpful definitions

- Pauli matrices/gates: $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

- Pauli- X eigenvectors: $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$.

- Hadamard gate: $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

- Phase gate: $S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$.

- CNOT gate: $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$.

For the grader only

Problem	1	2	3	4	5	6	Final Score
Score							